

The Fundamental Challenge

Why Automatic Differentiation Matters

The Problem:

Machine learning frameworks must solve a fundamental computational challenge: calculating derivatives through complex chains of mathematical operations efficiently and accurately.

Consider this complex function: $f(x) = x^2 \sin(x) e^{\cos(x)}$

~~Traditional Approach: Manual Calculus~~



175 billion parameters

ML model parameters are internal variables, such as weights and biases, that a model learns from training data to make predictions.

Even in this basic example, computing derivatives manually would require careful application of calculus rules. Now imagine scaling this to a neural network with millions of operations.

The Breakthrough Insight

Car dashboard analogy: Speedometer + Odometer



Traditional approach: checking odometer for position, speedometer for speed



Automatic differentiation: a futuristic dashboard that gives you both simultaneously

Position & Speed  Value & Derivative

Dual Numbers - The Secret Weapon

Dual(value, derivative)

$$x = \text{Dual}(2.0, 1.0)$$



Regular number:

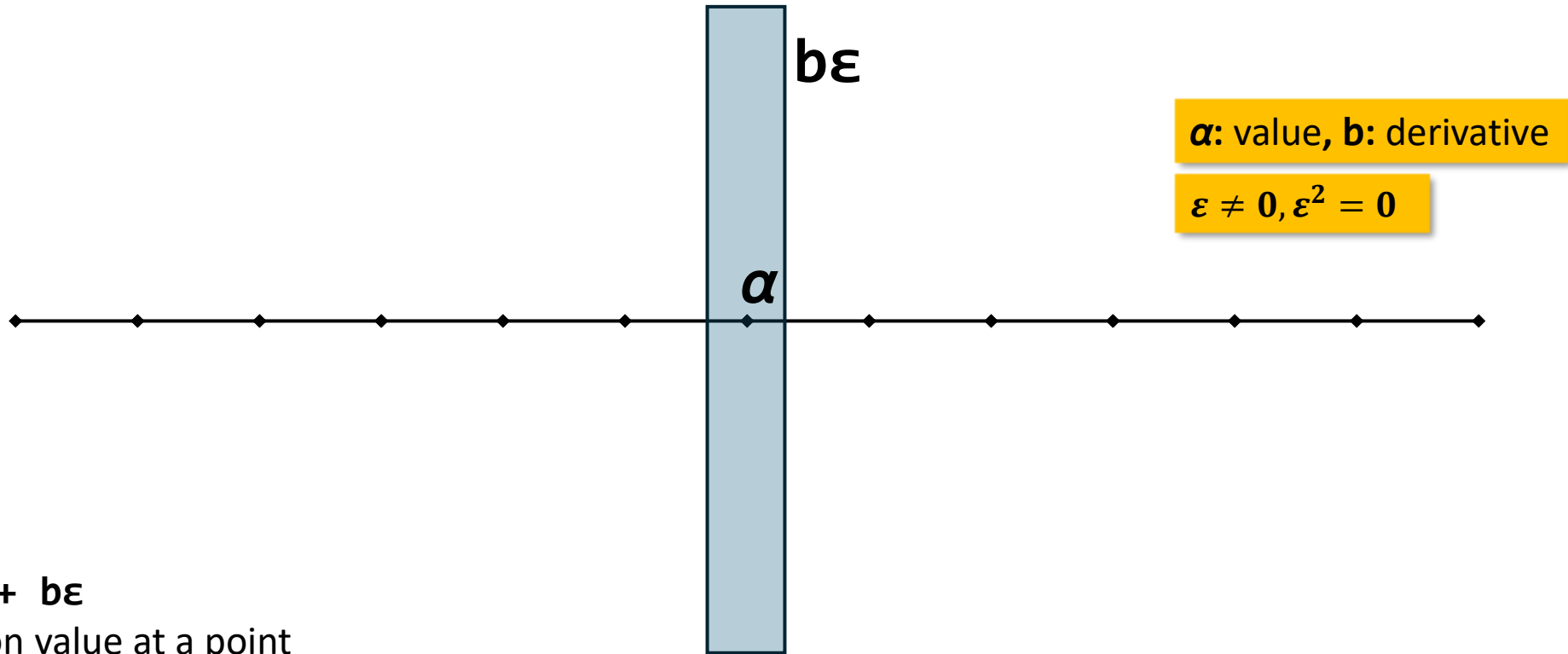
'I'm at position 5.'



Dual number

'I'm at position 5, moving at speed 2.'

What Are Dual Numbers?



- Form: $\alpha + b\varepsilon$
- α : function value at a point
- b : derivative at that point
- Defining property: $\varepsilon^2 = 0$ (nilpotent)

Nilpotent: a matrix or an algebraic element that, when multiplied by itself a sufficient number of times, results in a zero matrix or zero element, respectively

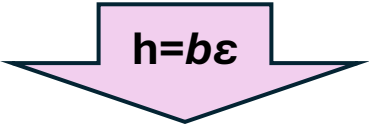
ϵ in Action: First-Order Taylor Expansion

The **Taylor series** of a function is an infinite sum of terms calculated from the values of its derivatives at a single point. It's an essential method for **function approximation**, numerical analysis, and solving differential equations.

Standard Taylor Expansion expansion around point a

(h is the increment from a to x , i.e. $h = x - a$)

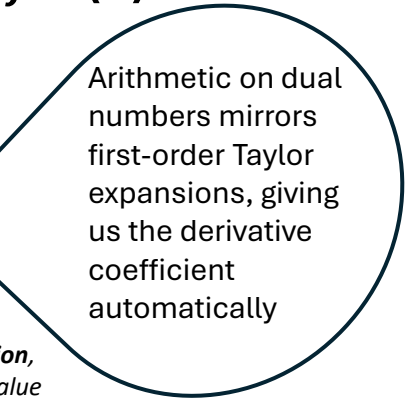
$$f(a + h) = f(a) + f'(a) \cdot h + f''(a) \cdot h^2/2! + f'''(a) \cdot h^3/3! + \dots$$


$$h = b\epsilon$$

$$f(a + b\epsilon) = f(a) + f'(a) \cdot b\epsilon$$

Automatic first-order expansion!

*The **first-order Taylor expansion**, or **linear approximation**, approximates a function $f(x)$ around a point a using its value and its first derivative at that point*



Arithmetic on dual numbers mirrors first-order Taylor expansions, giving us the derivative coefficient automatically